

Technical Report

Analysis of leakage from threaded portion of seal cooling pipe of MP-08 NDE side at WRPL, Viramgam (Western Region Pipeline)

Document No: TR-24-089

Nov. 2024



**Research and Development Centre
Indian Oil Corporation Ltd.
Sector-13, Faridabad, Haryana**

CONFIDENTIALITY

This document contains confidential information / data, proprietary to Indian Oil Corporation Ltd. Neither the information nor any part thereof may be duplicated or disclosed to anyone outside IOCL without the prior written consent of R&D Centre, Indian Oil Corporation Ltd. The Information and all portions thereof, including copies and derivative works thereof, shall, at the request of, and in accordance with instructions of R&D Centre, Indian Oil Corporation Ltd., be promptly returned to it, or destroyed.

Project Name : Analysis of leakage from threaded portion of seal cooling pipe of MP-08 NDE side at WRPL, Viramgam

Project Number : IOCL-R&D-AM-FA-077

Document Number : TR-24-089

Client : WRPL Viramgam
Indian Oil Corporation Limited

Service Provider : Research & Development Centre
Indian Oil Corporation Limited

Prepared	Reviewed	Approved
Gokul C Thakur	Dr. S P Singh	Mahendra Pal

Executive Summary

Viramgam is an important pump station of WRPL equipped with Mainline Pump Units (MLPUs) MP-07, MP-08, MP-09 for pumping towards Chaksu. During normal operation of VC-28" pipeline, a leakage was seen through CCTV in MP-08 Pump head through seal cooling pipe. Upon onsite investigation, a minor leakage was identified as originating from the thread portion of seal cooling pipe of MP-08 at NDE side. All the operational parameters including vibration history, flow rate, pressure and temperature trends of MP-08 (Table 1) were found to be within normal range.

Subsequently, seal cooling pipe of MP-08 was cut and sent to R&D center for assessment of cause of failure. A Detailed laboratory investigation was carried out on the received sample.

The thread roots (illustrative image - Fig. 18) are generally vulnerable locations for fatigue failure due to stress concentration effect at the root under dynamic loading conditions. The evidences collected in the present investigation amply indicate the failure of seal cooling pipe by fatigue failure mechanism. Further, the presence of inclusions (due to unclean steel used in manufacturing) and slightly higher hardness are likely to have induced brittleness in the pipe material and accentuated the fatigue crack growth.

Based on the detailed study, it can be recommended to:

- Keep the vibrations under specified limit.*
- Regular inspection (DPT/NDT) for presence of cracks especially near vulnerable locations (high SIF locations) such as threaded joints.*
- Use of clean steels to avoid manufacturing defects in rotary components*

Contents

Executive Summary	4
1.0 INTRODUCTION:	6
2.0 LABORATORY INVESTIGATIONS:	6
2.1 Visual Inspection:	6
2.2 Stereoscopic Examination:	7
2.3 Microscopic Examination:	8
2.4 Hardness Survey:	9
2.5 Positive Metal Identification (PMI):	9
2.6 Chemical Composition Analysis:	10
2.7 Fractographic Study:	10
2.8 Inclusion Rating:	12
3.0 RESULT AND DISCUSSIONS:	12
4.0 CONCLUSION:	12
5.0 RECOMMENDATIONS:	14
6.0 REFERENCES:	14

1.0 INTRODUCTION:

Viramgam is an important pump station of WRPL equipped with Mainline Pump Units (MLPUs) MP-07, MP-08, MP-09 for pumping towards Chaksu. During normal operation of VC-28" pipeline, a leakage was seen through CCTV in MP-08 Pump head through seal cooling pipe. Upon onsite investigation, a minor leakage was identified as originating from the thread portion of seal cooling pipe of MP-08 at NDE side. All the operational parameters including vibration history, flow rate, pressure and temperature trends of MP-08 (Table 1) were found to be within normal range.

Subsequently, seal cooling pipe of MP-08 was cut and sent to R&D center for assessment of cause of failure. A Detailed laboratory investigation was carried out on the received sample. The following enumerates the findings of the study.

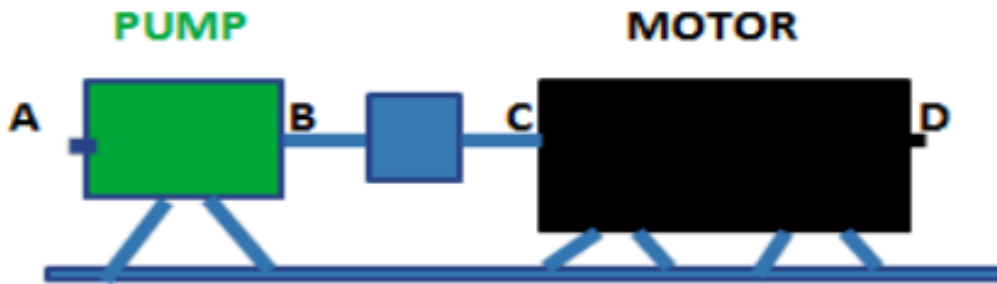


Fig. 1: A Schematic of Pump and Motor at a Pump Station

Table 1: Details of the failed Pump MP-08

Tag Name	VC28 MP-08 pump
Make	M/s Ruhrpumpen
Model	ZLM I 530 / 06
Diff head	313 m
Flow	1847 m3/hr.

2.0 LABORATORY INVESTIGATIONS:

2.1 Visual Inspection:

Visual examination revealed:

- No Significant corrosion was observed on the ID/OD surface of the sample.
- The failed surface of the sample revealed a fracture from near the thread end (Fig. 1).

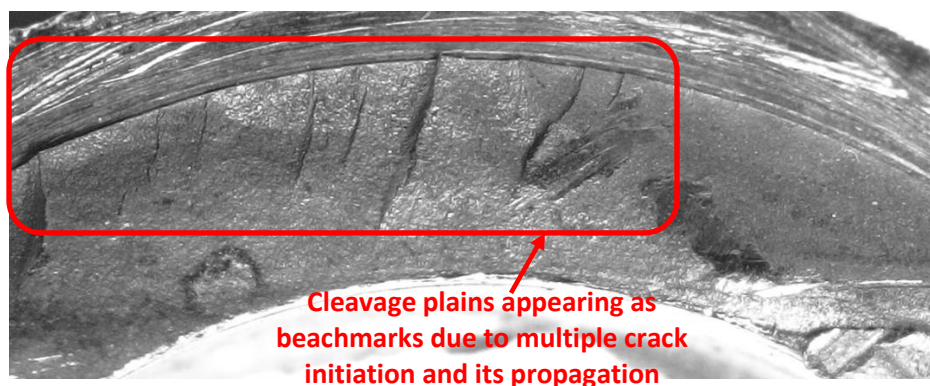
- Beach marks on the failed fracture surface could be observed from naked eye (Fig. 3).
- Thickness data revealed thickness around 3.5 mm.



Fig. 2-4: As received failed sample images showing failure from thread root and beach marks on the failed fracture surface

2.2 Stereoscopic Examination:

- Failed fracture surface of the sample was taken under stereoscope for identifying the features of the fracture surface.
- Stereoscopic images revealed symptoms of fatigue fractures such as crack initiation point, beach marks and propagation zones.



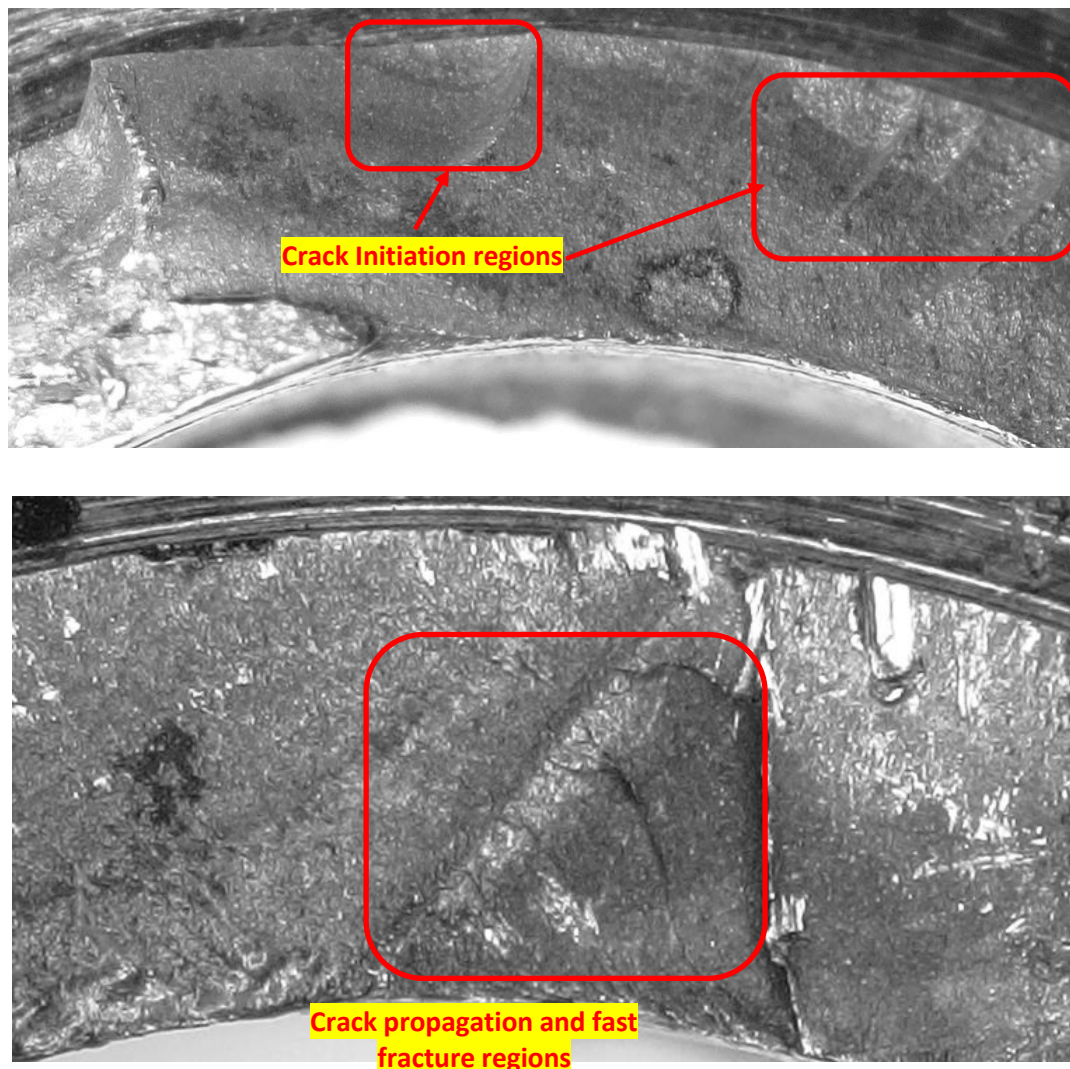


Fig. 5-7: Stereoscopic images showing the fatigue fracture features such as crack initiation, beach marks and propagation regions

2.3 Microscopic Examination:

Samples were cut from near to the failure location and its Circumferential-Thickness (C-T) section was prepared metallographically to mirror finish. Sample were further etched with 10 % hot oxalic acid solution for revelation of grain boundaries. Microscopic observations revealed:

- The presence of fine rounded inclusions was observed in unetched matrix (Fig. 8-9).
- Inclusions were uniformly distributed in the matrix.

- General Microstructure of austenitic nature with presence of twin boundaries typical of SS 316 grade structures (Fig. 10-11).
- Fine grain size of the austenite was observed in the matrix.

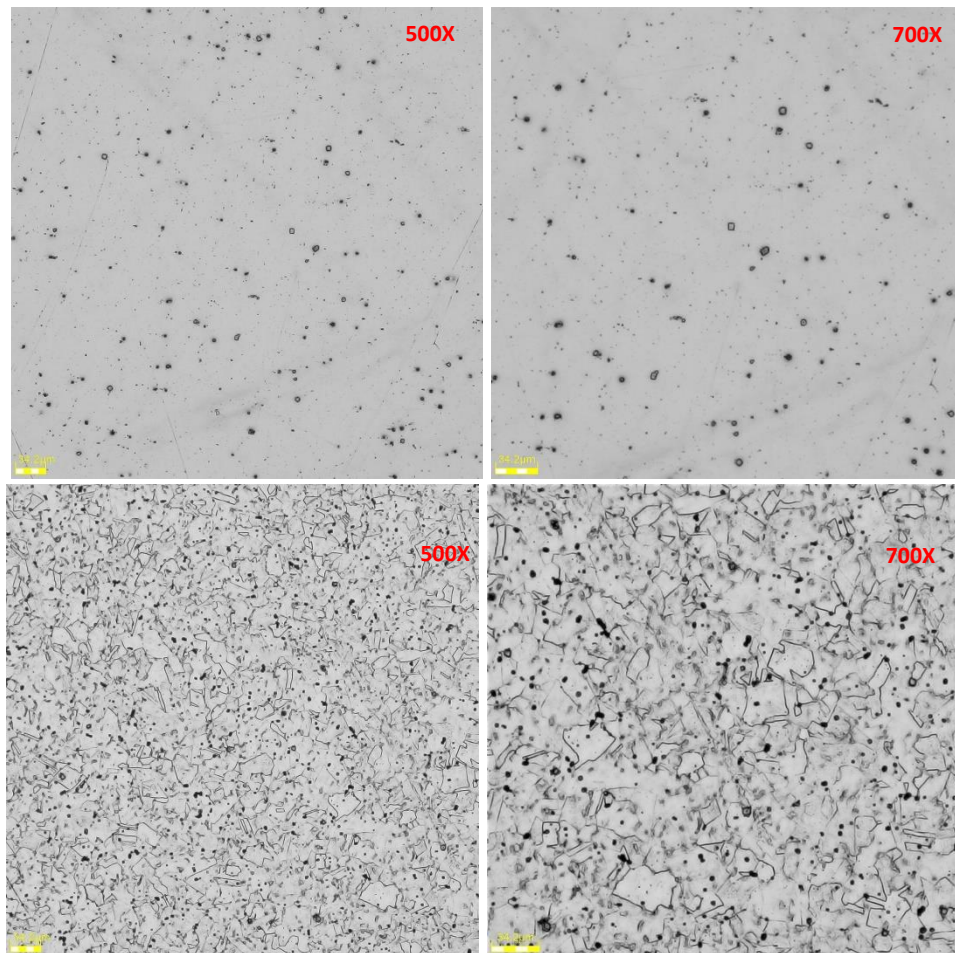


Fig. 8-11: Unetched and etched microstructures at different magnifications revealing inclusions

2.4 Hardness Survey:

The Bulk hardness examination of sample was performed on the Brinell scale. The Hardness survey was performed across the cross section of near and away from failure location samples at several locations. Minimum and maximum hardness were found to be 190 and 201 BHN respectively, which is slightly higher side of maximum specified limit of 192 BHN for SS316L grade steels.

2.5 Positive Metal Identification (PMI):

PMI of the samples were performed to identify the grade mismatch if any. PMI data revealed the material to be of SS 316 grades (Fig. 12).



Fig. 12: PMI results showing material to be of SS 316 grades

2.6 Chemical Composition Analysis:

Chemical composition analysis was carried out using spark spectroscopy method. Chemical composition results are given in table 2.

Table 2: Chemical composition results.

	C	Mn	P	S	Si	Cr	Ni	Mo
ASTM A 213	0.30	2.00	0.035	0.035	1.00	16-18	10-14	2-3
Sample	0.027	2.04	0.030	0.029	0.445	16.52	10.82	2.09

Chemical composition analysis results revealed overall the chemical conformity w.r.t. SS 316 grade.

2.7 Fractographic Study:

The failed fracture surface of the sample was taken under Scanning Electron Microscope (SEM) for analyzing the microscopic striation pattern, beach marks and modes of fracture. Fractographic study revealed:

- Fatigue induced beach marks along the fracture surface (C-T) surface.
- Stepped crack growth was witnessed due to merger of different crack growth planes.

- Fast fracture region was also seen along the fracture surface.

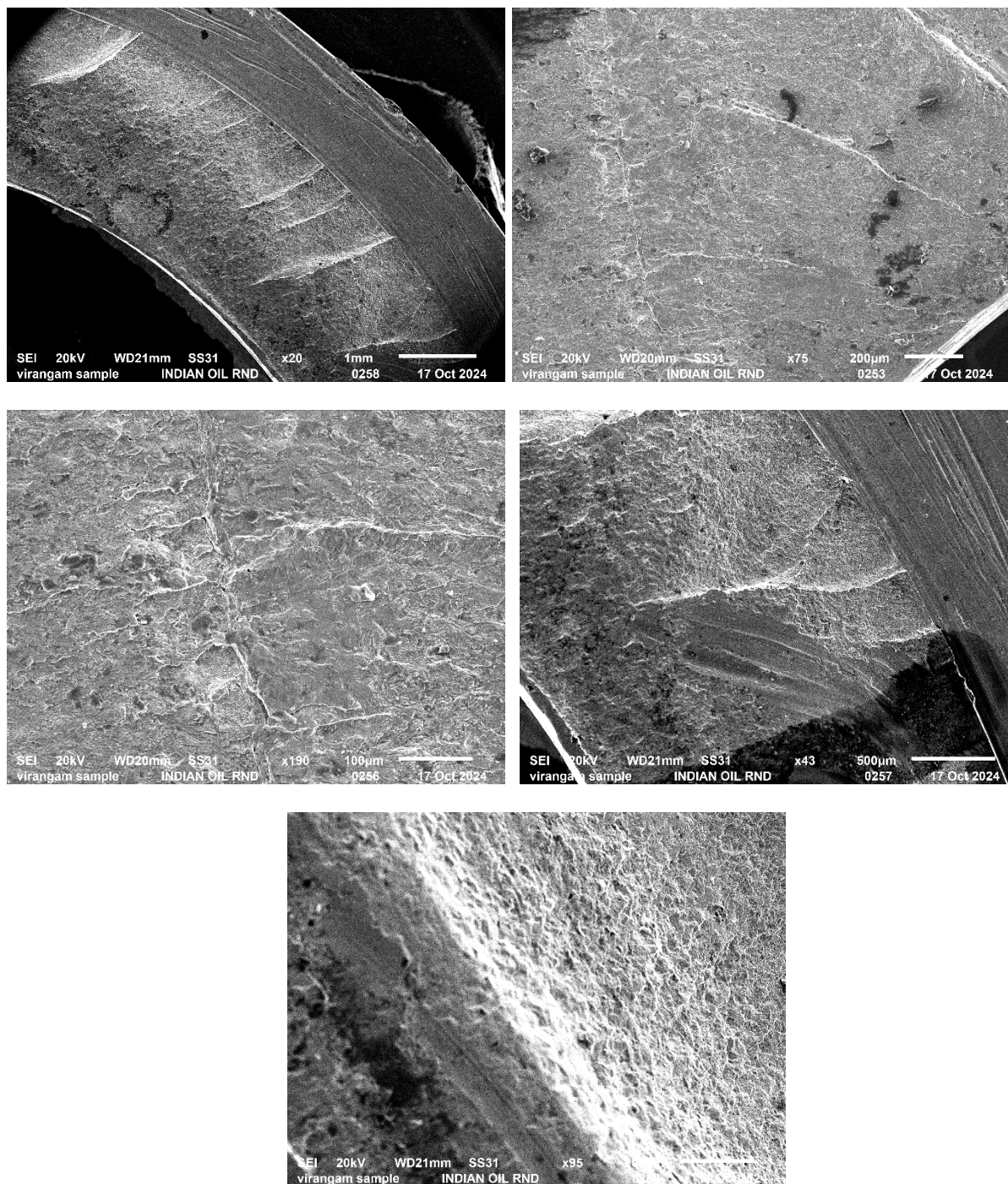


Fig. 13-17: Fracture surface showing fatigue striations and stepped crack growth

2.8 Inclusion Rating:

Inclusion rating of the samples as per ASTM E45-18a (2023) standard were also carried out for types of inclusions. Inclusion analysis revealed presence of Type "A" Thin-0.5 (Sulfide Type) and Type "D" Thin -0.5 (Globular Oxide Type) inclusions.

3.0 RESULT AND DISCUSSIONS:

- Visual examination revealed:
 - No Significant corrosion on the ID/OD surface of the sample.
 - Fracture from the thread root.
 - Beach marks on the failed fracture.
- Stereoscopic study revealed fatigue fractures such as crack initiation, beach marks and propagation zones.
- Microscopic observations revealed:
 - Presence of fine rounded inclusions uniformly distributed in the matrix.
 - General Microstructure of austenitic nature with presence of twin boundaries typical of SS 316 grade.
 - Fine grain size of the austenite in the matrix.
- The Hardness Study revealed Minimum and maximum hardness values 190 and 201 BHN respectively (slightly higher side of maximum specified limit of 192 BHN for SS316L grade steels).
- PMI and Chemical composition analysis results revealed overall the chemical conformity w.r.t. SS 316 grade.
- Fractographic study revealed
 - Fatigue beach marks along the fracture surface (C-T) surface.
 - Stepped crack growth was witnessed due to merger of different crack growth planes.
 - Fast fracture region was also seen along the fracture surface.

4.0 CONCLUSION:

The thread roots (illustrative image -Fig. 18) are generally vulnerable locations for fatigue failure due to stress concentration effect at the root under dynamic loading conditions. The evidence collected in the present investigation amply indicates the failure of seal cooling pipe by **Fatigue failure mechanism**. Further, the presence of inclusions (due to unclean steel used in manufacturing) and slightly higher hardness are likely to have

induced brittleness in the pipe material and accentuated the void nucleation, void coalescence, crack formation and its growth in stress axis (Fig. 19).

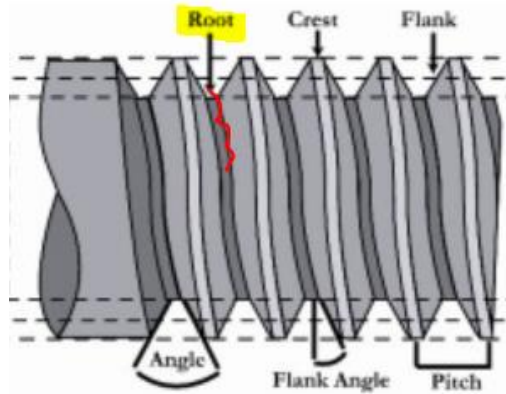


Fig. 18: Illustrative sketch showing the crack initiation from root of Threaded joint due to high SIF and its propagation in through thickness direction under fatigue

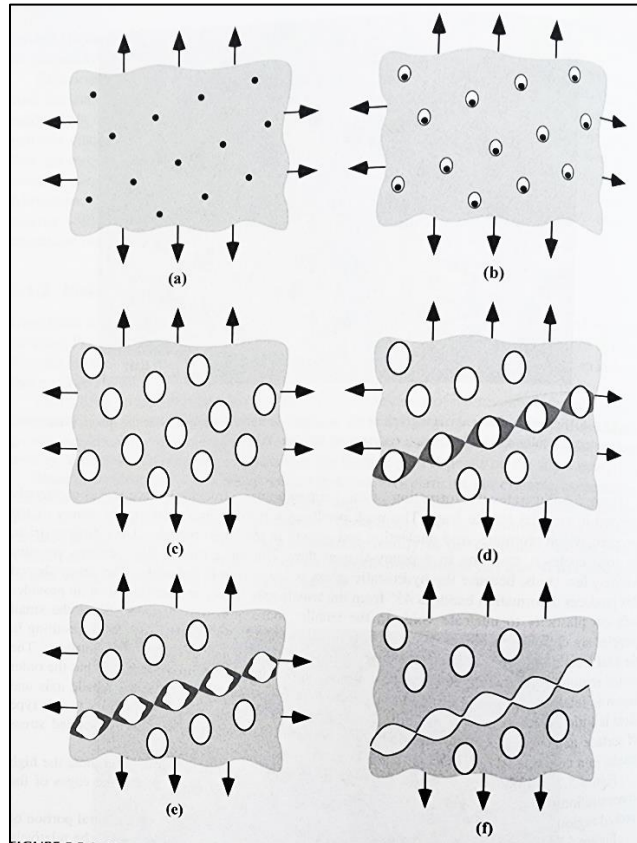


Fig. 19: Illustrative diagram showing role of inclusions in void nucleation, void coalescence, crack formation and its growth in stress axis [4]

5.0 RECOMMENDATIONS:

Based on the detailed study, it can be recommended to:

- Keep the vibrations under specified limit.
- Regular inspection (DPT/NDT) for presence of cracks especially near vulnerable locations (high SIF locations) such as threaded joints.
- Use of clean steels to avoid manufacturing defects in rotary components

6.0 REFERENCES:

- [1]. <https://www.eng-tips.com/viewthread.cfm?qid=204104> Recommended practice API 579, Damaged Mechanisms, Edition Dec. 2003.
- [2]. <https://nickelinstitute.org/media/4122/alloy-selection-for-service-in-sulphuric-acid-10057.pdf> Kareem Department of Mechanical Engineering, The Federal University of Technology, Nigeria.
- [3]. [https://www.researchgate.net/publication/276355005_CORROSION_EROSION_OF_316L_STAINLESS_STEEL_IN_SULFURIC_ACID_SOLUTION#:~:text=The%20mixture%20of%20gas%20and,a%20corrosion%20erosion%20damage%20pattern.5.Fatigue fracture expertise of train engine crankshafts A. Ktari , N. Haddar, H.F. Ayedi](https://www.researchgate.net/publication/276355005_CORROSION_EROSION_OF_316L_STAINLESS_STEEL_IN_SULFURIC_ACID_SOLUTION#:~:text=The%20mixture%20of%20gas%20and,a%20corrosion%20erosion%20damage%20pattern.5.Fatigue%20fracture%20expertise%20of%20train%20engine%20crankshafts,A.Ktari,N.Haddar,H.F.Ayedi)
- [4]. Fracture Mechanics : Fundamentals and Applications by T. L.. Anderson. 4th Edn